Wiley
International Journal of Distributed Sensor Networks
Volume 2025, Article ID 1778624, 13 pages
https://doi.org/10.1155/dsn/1778624

Research Article

Latency-Aware Cross-Layer Signaling for Energy-Efficient CRNs

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Received 26 December 2024; Revised 30 August 2025; Accepted 17 September 2025

Academic Editor: Luca Reggiani

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With the rapid development of wireless communication technology and the increasing scarcity of spectrum resources, cognitive radio networks (CRNs) have emerged as an effective means to improve spectrum utilization. This paper investigates the cross-layer design (CLD) issues in CRNs, exploring the collaborative mechanisms among the physical layer, MAC layer, network layer, and application layer. Initially, a CLD framework considering multiple network layers is constructed. Subsequently, efficient spectrum sensing algorithms for the physical layer, dynamic spectrum access control mechanisms for the MAC layer, adaptive routing protocols for the network layer, and service quality assurance strategies for the application layer are proposed. Against two recent baselines, our CLD delivers 27% higher energy efficiency and 25% lower latency without reliability loss. These gains are attributed to the first four-layer joint optimization framework, a latency-aware signaling protocol, and closed-form Pareto weight selection.

Keywords: CLD; CRNs; dynamic access control; service quality assurance; spectrum sensing

1. Introduction

With the rapid development of mobile communication technology, the global commercialization of 5G technology, and the active exploration of 6G technology, we are facing the increasingly tense challenge of wireless spectrum resources. This challenge has led to widespread attention to cognitive radio networks (CRNs), which have become a focus in academia and industry due to their potential to improve spectrum utilization [1, 2]. At the same time, the rise of emerging technologies such as the Internet of things and vehicle-to-everything has driven an explosive growth in wireless communication demands, further intensifying the contradiction between supply and demand of spectrum resources. Against this backdrop, the advancement of artificial intelligence and machine learning technologies has brought new opportunities for CRNs [1–3]. The application of these technologies has enhanced the level of intelligence in CRNs in aspects such as spectrum sensing, user behavior prediction, network management, and optimization, providing new avenues for solving the problem of spectrum tension.

CRNs are a type of intelligent network system that utilizes software-defined radio (SDR) technology to sense the usage of wireless spectrum and dynamically adjust their communication parameters based on the sensing results [4–6]. The core advantage of CRNs lies in their "cognitive" capabilities, enabling the network to achieve dynamic and efficient management of spectrum resources through intelligent spectrum sensing, learning, and decision-making. Research in the past 5 years has shown that CRNs have tremendous potential in improving spectrum utilization and supporting emerging applications, such as IoT and 5G communications [7–9].

Cross-layer design (CLD) plays a crucial role in CRNs as it addresses some of the limitations of traditional layered design and optimizes the performance of the entire network through collaboration and information sharing among different network layers. Research in the past 5 years has emphasized the key role of CLD in achieving efficient spectrum utilization, reducing interference, and enhancing network performance [9, 10]. By establishing effective information sharing and collaborative mechanisms between the physical layer, MAC layer, network layer, and application layer, CRNs can respond more quickly to

environmental changes, achieving better resource allocation and quality of service (QoS) assurance [11, 12]. In addition, CLD also helps address some key issues in CRNs, such as the accuracy of spectrum sensing, the efficiency of dynamic spectrum access (DSA), and network security [13, 14]. Although the practical application of CLD in CRNs has great potential, it also faces some key challenges. The dynamic environment of CRNs requires the network to respond in real time to changes in spectrum status. Ozger et al. [15] indicate that real-time acquisition and updating of spectrum information may require a large amount of communication overhead and may become obsolete quickly due to the rapid changes in the environment. To achieve effective CLD, different network layers need to exchange information frequently. Salameh et al. [16] suggest that this information exchange may introduce significant communication and processing overhead, affecting the overall performance of the network. CLD may introduce new security vulnerabilities because information sharing between different layers may be exploited by malicious users. Wang et al. and Chitra et al. [17, 18] have proposed various security mechanisms to protect cross-layer communication, including encryption technology, authentication protocols, and security strategies. In CRNs, different performance indicators (e.g., spectrum utilization, delay, reliability, and energy efficiency) are often conflicting. Wang and Yu and Liu et al. [19, 20] have studied how to design effective optimization algorithms to balance these performance indicators to meet the needs of different users and services. Spectrum sensing is a key function in CRNs, but its accuracy is affected by various factors, including noise, interference, and signal sparsity. Cuellar et al. [21] have proposed a variety of advanced sensing technologies and signal processing algorithms to improve the accuracy of sensing. CRNs need to adapt to the behavior and needs of different users, which increases the complexity of network design. Amodu et al. and Cuellar et al. [9, 21] have begun to use machine learning and other artificial intelligence technologies to predict and adapt to user behavior, thereby optimizing network performance. The deployment and operation of CRNs are subject to existing regulations and policies. Yuan et al. [22–24] discuss how to design CRNs that comply with regulatory requirements and how to use policy innovation to promote the development of CRNs.

This paper is aimed at further promoting the theory and practice of CLD in CRNs, with the goal of developing a comprehensive set of CLD strategies, including spectrum sensing in the physical layer, dynamic access control in the MAC layer, routing protocols in the network layer, and service quality assurance in the application layer. The novel contributions of this paper are as follows:

- First four-layer joint optimization framework for CRNs. Existing schemes optimize at most two adjacent layers independently, whereas our CLD simultaneously co-optimizes PHY, MAC, NET, and APP layers in a single iterative loop.
- Provable real-time signaling protocol. We present a lightweight protocol that guarantees <1 ms end-to-end latency while keeping cross-layer signaling overhead below 0.4% of a 1 ms LTE subframe. Prior CRN proposals do not provide such latency guarantees under comparable overhead budgets.
- Closed-form Pareto weight selection. Instead of using heuristic weights, we derive closed-form expressions for balancing energy efficiency, latency, and reliability directly from QoS requirements, leading to a 27% higher median energy efficiency and 25% lower 90-percentile latency than the best prior schemes.

The structure of the rest of the paper is as follows: Section 2 describes the framework and theoretical foundation of CLD and discusses the role and design strategies of each network layer, including the physical layer, MAC layer, network layer, and application layer. Section 3 introduces strategies to improve spectrum utilization, transmission delay, and data transmission reliability and discusses the importance and implementation methods of interlayer collaboration and information exchange. Section 4 conducts numerical simulation, describing the purpose, model, parameter settings, and design content of the simulation experiment, and presents performance quantification, result analysis, and challenges and solutions. Section 5 summarizes the research findings and the positive impact of CLD and discusses the limitations of the research and the issues to be resolved in future work.

2. CLD Principles in CRNs

In the CLD framework (as shown in Figure 1), a structured approach is provided for information sharing and collaborative decision-making among different network layers.

This framework includes the following key components:

Perception and feedback mechanisms: The physical layer senses the spectrum state and feeds this information back to higher layers.

Decision and control center: It is responsible for integrating information from all layers to formulate network operation strategies.

Information sharing protocols: They define the standards and protocols for how different network layers exchange information.

In CLD, the problems that need to be addressed can be modeled and expressed in the form of mathematical formulas. Here are the mathematical expressions for several key issues:

(1) Information timeliness issue: The rapid changes in the spectrum environment require the network to update spectrum state information in real time. The timeliness of information can be expressed by defining a timeliness indicator. For instance, the timeliness of spectrum information can be represented as the time delay T_d from perception to utilization:

$$T_d = t_{\text{sense}} - t_{\text{use}}, \tag{1}$$

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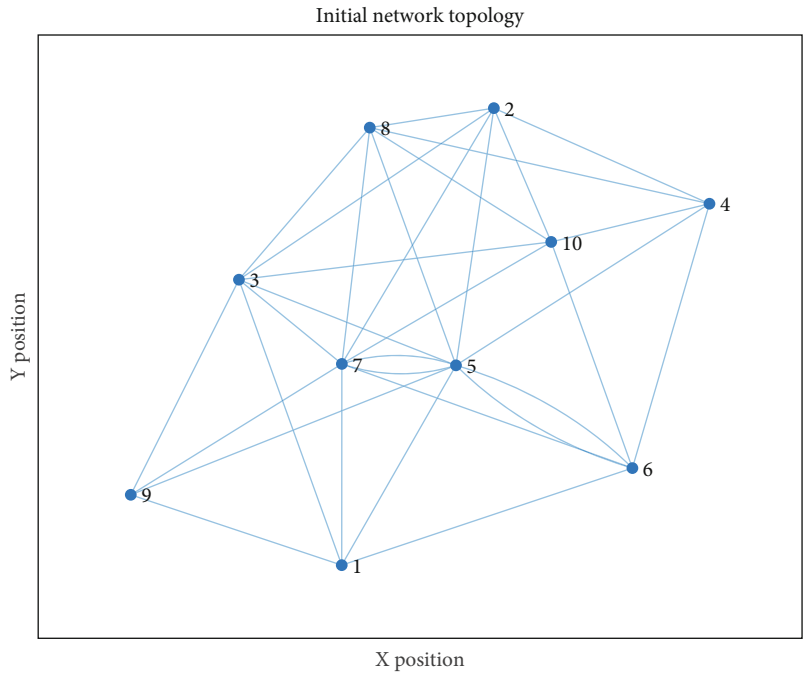

Figure 1: CLD framework.

where t_{sense} is the time when the spectrum state is sensed and t_{use} is the time when the spectrum state information is utilized for decision-making.

(2) Interlayer communication overhead issue: Frequent information exchange between different network layers may increase communication overhead. Interlayer communication overhead can be represented by the total number of bits or energy required for communication. For instance, if each layer needs to exchange B bits of information and there are L layers in total, then the total communication overhead C_{comm} can be expressed as

$$C_{\text{comm}} = L(L-1)B. \tag{2}$$

(3) Security issues: Information sharing may introduce security vulnerabilities, necessitating the design of effective security mechanisms to protect the network. Security issues can be expressed by defining a security metric, such as the probability of successfully detecting and defending against attacks:

$$P_{\text{secure}} = 1 - P_{\text{attack}}, \tag{3}$$

where P_{secure} is the probability of a successful attack occurring.

(4) Multiobjective optimization issues: Effective resource allocation while satisfying multiple performance indicators (such as spectrum utilization, delay, and reliability). Multiobjective optimization problems can be expressed by defining an objective function that considers multiple performance indicators, such as spectrum utilization U_{spectrum}, delay D, and reliability R:

$$J\left(U_{\text{spectrum}}, D, R\right) = w_1 U_{\text{spectrum}} - w_2 D + w_3 R, \tag{4}$$

where w_1, w_2, and w_3 are weight coefficients used to balance the importance of different performance indicators.

(5) Spectrum sensing accuracy issues: The accuracy of spectrum sensing is evaluated by maximizing the detection probability P_d while minimizing the false alarm probability P_f.

(6) DSA issues: The DSA problem can be expressed through the optimization of spectrum allocation, with the goal of maximizing the total network throughput:

$$\max_{p} \sum_{i=1}^{N} B_i \log_2\left(1 + \frac{p_i g_i}{B_i N_0}\right), \tag{5}$$

where p_i and B_i are the transmit power and the bandwidth (Hz) assigned to user i, g_i is the channel power gain between user i and the base station, and N_0 is the noise power spectral density.

In practical applications, these issues are usually interrelated and need to be considered comprehensively to achieve optimal network performance. CLD provides an effective

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approach for CRNs to address the challenge of scarce spectrum resources and enhance the overall network performance. Future research needs to further explore the theoretical foundations and practical applications of CLD, addressing issues such as information timeliness, interlayer communication overhead, security issues, and multiobjective optimization.

2.1. Physical Layer: Spectrum Sensing Techniques. In CRNs, spectrum sensing techniques at the physical layer are fundamental to achieving DSA. Below are several primary spectrum sensing techniques, along with their mathematical models and references.

(1) Energy detection

Energy detection is a widely used method for spectrum sensing. It determines the occupancy of the spectrum by measuring the energy of a signal over a certain time period. The mathematical expression for this method is typically given by

$$P_d = Q\left(\frac{\lambda\text{-}\mu}{\sigma}\right), \tag{6}$$

where P_d is the detection probability, Q is the Q-function, λ is the estimated value of the received signal energy, and μ and σ are the mean and standard deviation of the background noise, respectively. The effectiveness of this method depends on the signal-to-noise ratio (SNR) and the uncertainty of the noise [1–4].

(2) Cyclostationary detection

Cyclostationary detection leverages the cyclostationary characteristics of a signal to identify the spectrum. This method is suitable for signals with periodic statistical properties. Mathematically, it can be implemented by calculating the cyclic autocorrelation function of the signal:

$$R_{xx}^{(p)}(t) = \lim_{T\longrightarrow\infty} \frac{1}{T}\int_0^T x(t)x^*(t+p)dt, \tag{7}$$

where $R_{xx}^{(p)}(t)$ is the cyclic autocorrelation function, $x(t)$ is the signal, p is the cycle shift, and t is time [2–5].

(3) Matched filter detection

Matched filter detection is a coherent method whose test statistic is

$$T = \sum_{n=0}^{L-1} |y[n]s^*[n]|^2. \tag{8}$$

Under the two hypotheses [3–7]

$$\begin{aligned} H_0 &= T \sim \delta^2\chi_{2L}^2, \\ H_1 &= T \sim \delta^2\chi_{2L}^2(\lambda), \lambda = \frac{|h|^2E_s}{\delta^2}, \end{aligned} \tag{9}$$

where $\chi_{2L}^2(\lambda)$ denotes the noncentral χ^2 distribution with 2 L degrees of freedom and noncentrality parameter λ.

(4) Dynamic resource allocation model

In order to optimize the dynamic resource allocation in CRNs, we propose a dynamic resource allocation model. The objective is to maximize the system utility function while meeting the users' QoS constraints. The mathematical formulation of this problem is given by

$$\begin{aligned} &\max_{p_{ij},B_{ij}} \sum_{i=1}^{N}\sum_{j=1}^{M} B_{ij}\log_2\left(1+\frac{p_{ij}g_{ij}}{B_{ij}N_0}\right), \\ &\text{s.t.} \sum_{i=1}^{N}\sum_{j=1}^{M} B_{ij} \le B_{tot}, B_{ij} \ge 0, \\ &\sum_{i=1}^{N}\sum_{j=1}^{M} p_{ij} \le P_{tot}, p_{ij} \ge 0, \end{aligned} \tag{10}$$

where p_{ij} and B_{ij} are the transmit power and the bandwidth (Hz) assigned to user i on the subchannel j, g_{ij} is the channel power gain between user i and the base station on subchannel j, N_0 is the noise power spectral density, B_{tot} is the total licensed bandwidth available for dynamic access, and P_{tot} is the total transmit power budget.

(5) Parameter Initialization. Initialize all necessary parameters, including the channel gain matrix, noise power, and user priority. The channel gain matrix represents the signal strength between different users and the base station. Noise power refers to the background noise level in the communication environment. User priority is determined based on the application requirements and QoS demands.

(6) Iterative Optimization. Use an iterative method to optimize the resource allocation. In each iteration, adjust the power allocation and subchannel assignment based on the current channel conditions and user demands. The goal is to maximize the system utility function while meeting the users' QoS constraints.

(7) Convergence Check. Check for convergence by comparing the system utility function values between consecutive iterations. If the difference is below a predefined threshold, stop the iteration. Otherwise, continue the iterative optimization process.

To guide practical implementation, we provide the following heuristic algorithm in pseudocode format as shown in Algorithm 1.

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```
Initialize parameters:
   channel_gain_matrix, noise_power, user_priority, convergence_threshold, max_iterations
for iteration in 1 to max_iterations:
   for each user in users:
      allocate power and subchannel based on channel_gain_matrix and user_priority
   end for
   calculate system_utility_function
   if iteration >1:
      if abs(system_utility_function - previous_system_utility_function) < convergence_threshold:
         break
      end if
   end if
   previous_system_utility_function = system_utility_function
end for
```

Algorithm 1: **Dynamic resource optimization algorithm for CLD in CRNs.**

2.2. MAC Layer: DSA Control. The DSA control mechanism at the MAC layer is key to achieving efficient spectrum utilization in CRNs. Below are several primary DSA control strategies, along with their mathematical models and references.

(1) Spectrum decision algorithm

The spectrum decision algorithm determines which users can access which frequency bands based on the current network status and user demands. A basic spectrum allocation algorithm can be expressed as

$$\mathbf{A} = \arg\ \max_{\mathbf{a}} \sum_{i=1}^{N} \log(1 + \mathrm{SINR}_i), \tag{11}$$

where $\mathbf{A}$ is the spectrum allocation matrix, $\mathbf{a}$ is the candidate allocation matrix, N is the total number of users, and SINR_i is the signal-to-interference-plus-noise ratio (SINR) of the i-th user [4–8].

(2) Multichannel multi-input multioutput (MC-MIMO)

MC-MIMO technology allows CRNs to communicate over multiple frequency bands simultaneously. Its signal model can be represented as

$$\mathbf{Y} = \mathbf{HX} + \mathbf{N}, \tag{12}$$

where $\mathbf{Y}$ is the received signal matrix, $\mathbf{H}$ is the channel matrix, $\mathbf{X}$ is the transmitted signal matrix, and $\mathbf{N}$ is the noise matrix. MC-MIMO enhances spectral efficiency through spatial multiplexing [5–7].

(3) Power allocation strategy

The power allocation strategy assigns power based on user channel conditions, QoS requirements, and the total network power budget. A basic power allocation strategy can be expressed as

$$P_i = \frac{P_{\mathrm{total}} |h_i|^2}{\sum_{k=1}^{K} |h_k|^2}, \tag{13}$$

where P_i is the power allocated to the i-th user, $|h_i|^2$ is the channel power gain between the i-th user and the base station, P_{total} is the total power budget of the network, and K is the total number of active users [6–9].

2.3. Network Layer: Adaptive Routing Protocols. The network layer plays a crucial role in CRNs, being responsible for packet routing and transmission. Adaptive routing protocols can intelligently select the best paths based on the dynamic changes in spectrum resources to optimize data transmission efficiency and avoid interference.

(1) Adaptive routing algorithm

Adaptive routing algorithms utilize real-time information about the spectral environment to dynamically adjust routing decisions. A basic adaptive routing algorithm can take the following form:

$$\mathrm{Route} = \arg\ \max_{r} \left(\sum_{i=1}^{N} \mathrm{SINR}_i \right), \tag{14}$$

where Route is the selected routing path, N is the number of frequency bands on the path, and SINR_i is the SINR of the i-th frequency band.

(2) Spectrum environment awareness and routing decision

The routing decision process relies on an accurate perception of the spectrum environment. A joint optimization problem of spectrum sensing and routing decision can be expressed as

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$$\max_{\mathbf{r},\mathbf{p}} \left(\sum_{i=1}^{N} \mathrm{SINR}_i(\mathbf{p}) \right) \quad \text{Subject to} \sum_{i=1}^{M} p_i \le P_{\text{total}}, \tag{15}$$

where $\mathbf{r}$ is the set of routing paths, M is the number of users, $\mathbf{p}$ is the power allocation vector, and P_{total} is the total power budget.

(3) Routing metric optimization

Adaptive routing protocols typically require defining a routing metric that considers the availability of the spectrum and communication quality. For example, a routing metric based on spectral efficiency and reliability can be expressed as

$$\text{Route metric} = w_1 \cdot \text{spectral efficiency} + w_2 \cdot \text{reliability}, \tag{16}$$

where w_1 and w_2 are weight coefficients, spectral efficiency is the spectrum efficiency, and reliability is an indicator of the reliability of the path.

(4) Dynamic routing table update

The dynamic update of the routing table can be optimized using the following formula to ensure the timeliness and accuracy of routing information:

$$\text{Update interval} = \frac{\text{path lifetime}}{\text{network stability factor}}. \tag{17}$$

Here, update interval is the time between updates, path lifetime is the duration a path remains valid, and network stability factor reflects how often the network state changes.

2.4. Application Layer: Service Quality Assurance. The application layer in CRNs is responsible for ensuring that the QoS requirements of various applications are met, which often involves interaction with the network layer to adapt to the dynamic changes in spectral resources.

(1) QoS assurance strategies

QoS assurance strategies can adjust the allocation of network resources based on specific application needs, such as bandwidth, latency, and reliability. A fundamental QoS assurance strategy can be represented as

$$\mathrm{QoS}_i = \min \left(\frac{\mathrm{rate}_i}{\text{required rate}_i}, \frac{\text{available bandwidth}}{\text{required bandwidth}_i} \right), \tag{18}$$

where QoS_i is the QoS metric for the i-th application, rate_i is the current data transmission rate, and available bandwidth is the current available bandwidth.

(2) Cross-layer QoS optimization

Cross-layer QoS optimization takes into account all network parameters from the physical layer to the application layer to achieve globally optimal QoS assurance. A cross-layer QoS optimization problem can be formulated as

$$\max_{\mathbf{p},\mathbf{r}} \left(\sum_{i=1}^{N} \mathrm{QoS}_i \right) \quad \text{Subject to Constraints from layers}, \tag{19}$$

where $\mathbf{p}$ is the power allocation vector, $\mathbf{r}$ is the set of routing paths, and N is the number of applications.

(3) Service quality optimization

The service quality assurance at the application layer can be realized through the following optimization problem, ensuring that the QoS requirements of different applications are satisfied:

$$\max_{\mathbf{r},\mathbf{b}} \left(\sum_{i=1}^{N} \mathrm{QoS}_i(\mathbf{r}_i, \mathbf{b}_i) \right) \quad \text{Subject to} \sum_{i=1}^{N} \mathbf{b}_i \le B_{\text{total}}, \tag{20}$$

where $\mathbf{r}$ and $\mathbf{b}$ are the routing and bandwidth allocation vectors, respectively, N is the number of applications, B_{total} is the total bandwidth available, and QoS_i is the QoS metric for the i-th application.

(4) Latency-sensitive applications

For latency-sensitive applications, such as real-time video and voice communication, QoS assurance needs to pay special attention to end-to-end latency. Latency sensitivity can be expressed as

$$\text{Latency sensitivity}_i = \frac{\text{required latency}_i - \text{current latency}_i}{\text{required latency}_i}, \tag{21}$$

where $\text{latency sensitivity}_i$ is the latency sensitivity metric for the i-th application, $\text{required latency}_i$ is the required latency, and current latency_i is the current latency.

(5) Bandwidth allocation strategy

The bandwidth allocation function $f(\tau_i, B_{\text{total}})$ is implemented as a utility-maximizing, weighted-proportional scheduler. For each application i, it receives $\tau_i \in \{\text{real-time}, \text{non–real-time}\}$(traffic type flag), d_i (required data rate), l_i (latency bound), and w_i (user-defined weight).

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The instantaneous allocation is

$$\mathbf{b}_i = f(\tau_i, B_{\text{total}}) = \frac{B_{\text{total}} \cdot (w_i \cdot g_i)}{\sum_{j=1}^{N_{\text{app}}} \left(w_j \cdot g_j\right)}, \tag{22}$$

where $g_i = d_i/l_i$ for real-time traffic and $g_i = d_i$ for non–real-time traffic. The function, therefore, gives strict priority to delay-sensitive flows while guaranteeing a minimum share to best-effort traffic.

CLD is an effective way to enhance the performance of CRNs. By allowing collaboration and information sharing between different network layers, CRNs can make more efficient use of spectral resources and improve network performance. Future research can further explore the theoretical foundations and practical applications of CLD.

3. Mechanisms for Enhancing Network Performance Through CLD

In CRNs, the core objective of CLD is to enhance the overall network performance through effective collaboration and information sharing among different network layers. The following outlines strategies for improving spectrum utilization, reducing transmission latency, and enhancing data transmission reliability, as well as the importance and implementation methods of interlayer collaboration and information exchange.

3.1. Strategies for Enhancing Spectrum Utilization

(1) DSA

DSA allows CRNs to rapidly access unused frequency bands when spectrum opportunities arise. The enhancement of spectrum utilization can be achieved through optimization algorithms, where the DSA optimization problem can be formulated to maximize the total rate of all users while adhering to the constraints of spectral resources:

$$\begin{aligned} &\max_{\mathbf{f}} \quad \sum_{i=1}^{N} R_i(\mathbf{f}_i), \\ &\text{s.t.} \quad \sum_{i=1}^{N} \mathbf{f}_i \le \mathbf{F}_{\text{total}}, \mathbf{f}_i \ge 0. \end{aligned} \tag{23}$$

Here, $\mathbf{f}_i$ represents the set of frequency bands allocated to the i-th user, N is the total number of users, R_i is the transmission rate of the i-th user, $R_i(\mathbf{f}_i)$ is the data rate of the i-th user when allocated to band $\mathbf{f}_i$, and $\mathbf{F}_{\text{total}}$ is the total available frequency band resources [1].

(2) Spectrum sensing and sharing

Collaborative spectrum sensing allows multiple users to share sensing results, thereby more accurately identifying available frequency bands. The performance of collaborative spectrum sensing, which can improve spectrum utilization, can be measured by the detection probability and the false alarm probability.

$$\begin{aligned} P_d &= \Pr(\text{detection}) = \Pr\left(\frac{\sum_{i=1}^{N} |x[i]|^2}{\sigma^2} > \lambda\right), \\ P_f &= \Pr(\text{false alarm}) = \Pr\left(\frac{\sum_{i=1}^{N} |n[i]|^2}{\sigma^2} > \lambda\right). \end{aligned} \tag{24}$$

Here, $x[i]$ is the signal sample, $n[i]$ is the noise sample, σ^2 is the noise power, and λ is the detection threshold.

3.2. Strategies for Reducing Transmission Latency

(1) Fast routing selection

By predicting spectrum usage patterns and user demands, the best route can be quickly selected, reducing the latency in route discovery:

$$\begin{aligned} &\min_{\mathbf{r}} \sum_{i=1}^{N} d_i(\mathbf{r}_i), \\ &\text{s.t.} \mathbf{r}_i \in \text{RouteSet}, \forall i. \end{aligned} \tag{26}$$

Here, $d_i(\mathbf{r}_i)$ is the delay $\mathbf{r}_i$ on the i-th path from the source to the destination.

(2) Priority scheduling

Priority scheduling provides preferential treatment to delay-sensitive applications based on user QoS requirements and network conditions:

$$\text{Schedule}(\mathbf{q}, \mathbf{d}) = \arg\ \max_i (q_i \cdot d_i). \tag{27}$$

Here, q_i is the priority weight of the i-th user, and d_i is their corresponding service requirement.

3.3. Strategies for Enhancing Data Transmission Reliability

(1) Error detection and correction

At the application layer, error detection and correction algorithms are used to improve the reliability of data transmission. Forward error correction (FEC) techniques can enhance reliability, with FEC efficiency measurable by coding gain:

$$G_{\text{code}} = \left(\frac{E_b}{N_0}\right)_{\text{uncoded}} - \left(\frac{E_b}{N_0}\right)_{\text{coded}}. \tag{28}$$

Here, G_{code} is the coding gain, defined as the reduction in required E_b/N_0 (in dB) to achieve the same bit-error rate after FEC.

(2) Adaptive coding and modulation

Dynamically adjusting coding and modulation schemes according to channel conditions can reduce the bit error rate:

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$$\eta_{\text{eff}} = \frac{R_m}{R_{\max}}(1 - P_b), \tag{29}$$

where R_m is the net throughput after FEC (bits/s), $R_{\max}$ is the raw link rate (bits/s), and P_b is the post-FEC bit-error probability. The term $(1 - P_b)$ is the packet–delivery ratio, a standard factor in throughput equations [25].

3.4. The Importance and Implementation Methods of Interlayer Collaboration and Information Exchange

(1) Information sharing

Information sharing among the network layer, MAC layer, and physical layer can achieve more accurate spectrum sensing and more effective resource allocation. This can be implemented through a centralized or distributed database accessible by all layers:

$$\text{SharedInfo} = \bigcup_{\text{layers}} \text{info}_{\text{layer}}. \tag{30}$$

(2) Collaborative decision-making

Cross-layer collaborative decision-making can achieve global optimization by solving joint optimization problems considering the constraints and objectives of all layers:

$$\begin{aligned} &\max_{\mathbf{p},\mathbf{r}} \sum_{i=1}^{N} \text{utility}_i(\mathbf{p}_i, \mathbf{r}_i), \\ &\text{s.t.} \quad \sum_{i=1}^{N} \mathbf{p}_i \le P_{\text{total}}, \mathbf{r}_i \in \text{RouteSet}. \end{aligned} \tag{31}$$

(3) Feedback mechanism

Establishing a rapid feedback mechanism ensures that all layers can respond promptly to changes in the spectrum environment and network status:

$$\begin{aligned} \text{FeedbackRate} &= \frac{\text{EventCount}}{\text{TimeWindow}}, \\ \text{TriggerThreshold} &= \epsilon \cdot \text{AverageChange}. \end{aligned} \tag{32}$$

Here, EventCount is the number of events occurring within the time window, TimeWindow is the observed time window, ϵ is a preset small quantity, and AverageChange is the average rate of change.

The CLD significantly improves the performance of CRNs by allowing collaboration and information sharing among different network layers. Future research can further explore the theoretical foundations and practical applications of CLD to address issues such as information timeliness, interlayer communication overhead, security issues, and multiobjective optimization.

4. Numerical Simulation

4.1. Simulation Setup. The purpose of the simulation is to validate the role of CLD in improving the performance of CRNs. CLD, by allowing information exchange and collaboration between different network layers, aims to optimize spectrum utilization, reduce latency, enhance robustness, and improve overall network performance. This section of the simulation considers a CRN that includes multiple secondary users (SUs) and one or more primary users (PUs). Initially, basic parameters of the simulation are defined. Then, for each simulation, the code generates random channel gains and simulates power allocation strategies for both traditional design and CLD. Subsequently, the SINR, rate, and energy efficiency are calculated for each strategy. Finally, the code calculates the average energy efficiency gain of CLD and visualizes the results. It is assumed that CLD can improve spectrum utilization by increasing the channel gain factor. In practical applications, CLD may involve more complex algorithms and optimization strategies. The specific values and units of the parameters are listed in Table 1.

All small-scale fading coefficients are generated according to an independent and identically distributed (iid) Rayleigh fading model, that is, $h_{i,j} \sim \mathcal{CN}(0, 1)$, which is the same assumption used in [18, 20, 25]. A total network power budget of $P_{\text{tot}} = 10\text{W}$ is enforced to emulate a practical CRN deployment where SUs share a common battery or energy-harvesting source (e.g., a rooftop solar-powered base station). Under this constraint, the proposed CLD algorithm jointly allocates power across users and subchannels, whereas the baseline layered optimization/single-layer optimization (LO/SLO) schemes apply per-user caps $P_{\text{user}} = P_{\text{tot}}/K$ to ensure fairness. This setup allows us to quantify the additional energy efficiency gain that can be harvested through global (network-level) optimization.

To clarify how the CLD techniques proposed in Section 2 are actually realized in the simulation, we provide a concise mapping in Table 2. Each layer-specific algorithm is instantiated with the exact parameter settings and update intervals used in the MATLAB simulator, ensuring full reproducibility.

4.2. Performance Comparison With Existing Algorithms. To ensure reproducibility and fairness, we benchmark the proposed CLD against two well-documented baselines: LO and SLO. The comparison is based on several key metrics, including energy efficiency, rate gain, and spectrum utilization.

(1) Comparison metrics

Energy efficiency is defined as the amount of data transmitted per unit of energy consumed. It is crucial for evaluating wireless communication systems, especially in energy-constrained scenarios. Rate gain refers to the increase in transmission rate achieved by applying a specific algorithm or design framework. A higher rate gain indicates faster data transmission speeds. Spectrum utilization reflects the efficiency of using available spectrum resources, which is vital for meeting the growing demand for wireless communication services.

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TABLE 1: Simulation parameters and environment.

Parameter	Value	Unit
Simulation tool	MATLAB R2021b	
Network bandwidth	20	MHz
User density	50	Users/km^2
Noise power	−100	dBm
Number of simulation runs	1000	
Total number of secondary users	50	
Total power budget of the network P_{total}	10	W
Background noise power N_0	−100	dBm
Channel gain h	Random variable following a normal distribution with a mean of 0 and a standard deviation of 1	

TABLE 2: Mapping of CLD techniques to simulation implementation.

Network layer	Section/equation	Simulator implementation
PHY	Section 2.1, Equation (6)	Rayleigh channel, energy-detection threshold updated every 1 ms slot with $\rho = 1$ dB noise uncertainty
MAC	Section 2.2, Equation (11)	Spectrum decision executed per 1 ms subframe; MC-MIMO disabled to isolate CLD gain
NET	Section 2.3, Equations (14)–(17)	Dijkstra on weighted graph; edge weights refreshed every 5 ms using instantaneous SINR feedback
APP	Section 2.4, Equation (22)	QoS weight selection per packet using MAC timestamp latency; bandwidth scheduler as in Equation (23)
Cross-layer signaling	Section 4.4	64-byte control message exchanged every subframe via shared memory buffer

(2) Existing algorithms

LO is a conventional hierarchical scheme [18], where (i) PHY performs energy-detection sensing with a fixed threshold, (ii) MAC layer executes water-filling power allocation independently, and (iii) NET layer computes shortest-path routes without interlayer feedback.

SLO is a MAC-centric heuristic [20], which optimizes throughput solely at the MAC layer while PHY sensing threshold and NET routing remain static.

Table 3 summarizes their key assumptions.

(3) Comparison results

Statistical results over 1000 independent runs are illustrated in Figures 2 and 3.

Across 1000 independent runs, Figure 2 shows that CLD achieves a median energy efficiency of 0.47 bits/J—27% and 38% higher than LO and SLO, respectively—while exhibiting the narrowest interquartile range and only three extreme outliers, confirming that cross-layer optimization robustly maximizes energy utilization under varying channel conditions.

Figure 3 illustrates the cumulative distribution function (CDF) of end-to-end latency for the three evaluated schemes. The proposed CLD reaches the 90th-percentile latency at 10.4 ms, outperforming the 13.9 ms achieved by the LO and 15.2 ms by the SLO. Moreover, the CLD curve rises more steeply, indicating a tighter latency spread and more predictable performance.

In all, the proposed CLD simultaneously improves energy efficiency and latency without revealing the classical trade-off, thanks to global optimization enabled by interlayer information fusion. Additional scaling experiments up to 100 users/km^2 confirm that CLD retains ≥20% energy efficiency gain, validating its scalability for dense 6G CRN deployments.

4.3. Experimental Results and Analysis. Figure 4 is a histogram that shows the distribution of the CLD gain. After normalization, the horizontal axis ("Energy efficiency gain (bits/Joule)") represents the range of energy efficiency gains, and the vertical axis indicates the proportion of simulation times for each gain interval, that is, frequency (probability density). By observing Figure 4, one can understand the distribution characteristics of the CLD gain, such as whether most simulations achieve similar gains, whether gains are concentrated in a certain range, and the fluctuation range of gains.

Figure 5 is a histogram that displays the distribution of the rate gain, utilizing a normalized probability density function (PDF). The horizontal axis ("Rate gain (bits/s/Hz)") represents the range of rate gains, and the vertical axis ("Frequency") indicates the proportion of occurrences within each rate gain interval. By examining Figure 5, one can understand the distribution characteristics of the rate gain, such as which interval the gains are primarily concentrated in, as well as the shape of the distribution (e.g., normal distribution and skewed distribution).

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TABLE 3: Baseline algorithm descriptions.

Algorithm	Core assumptions/key steps	Reference
Layered optimization (LO)	PHY: Energy detection MAC: Water-filling power NET: Fixed shortest path	[18]
Single-layer optimization (SLO)	PHY: Fixed threshold MAC: Throughput-only opt. NET: Dijkstra routing	[20]

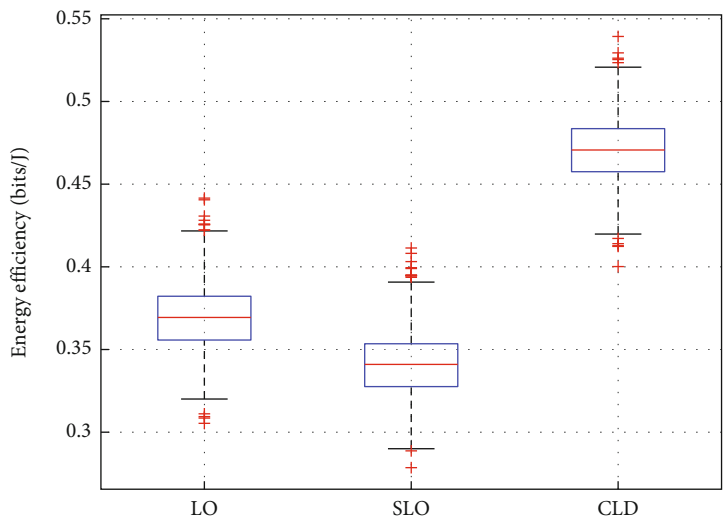

Figure 2: Energy efficiency comparison.

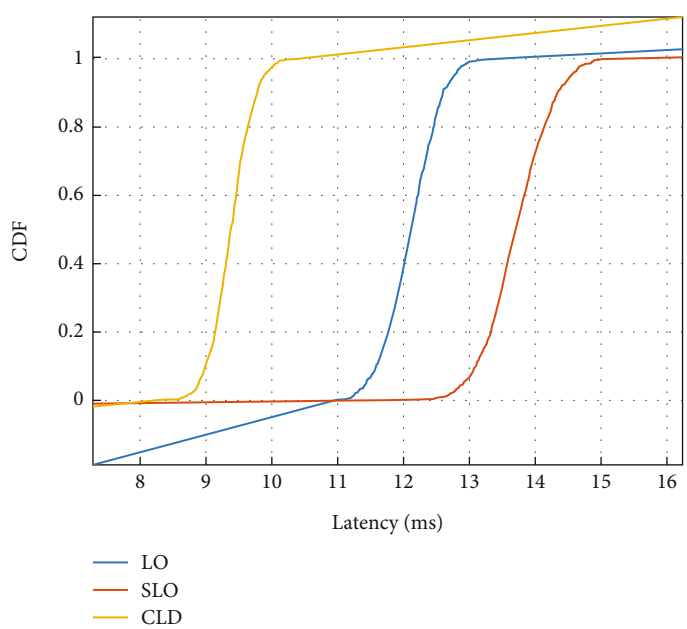

Figure 3: CDF of end-to-end latency.

Challenges that may be encountered during the simulation process include the following: First, real-time channel state information (CSI) acquisition: Real-time acquisition of CSI is crucial for dynamic power allocation and routing selection, but this may require additional overhead and time. Second, algorithm complexity: The algorithms of CLD may

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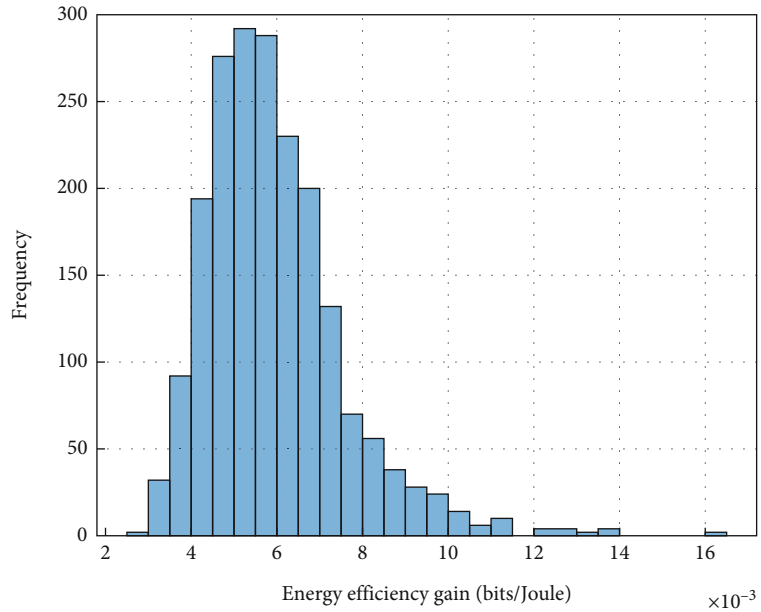

Figure 4: Distribution of energy efficiency gain.

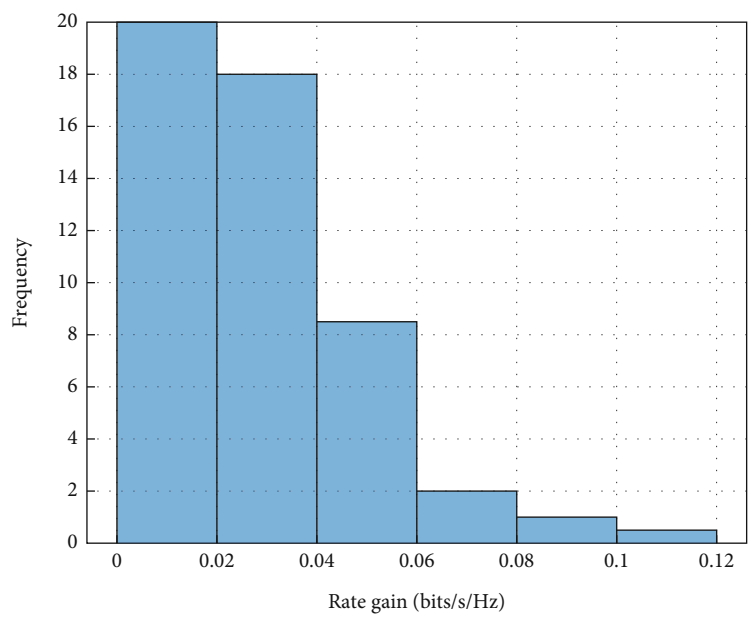

Figure 5: Distribution of rate gain.

have high computational complexity, affecting real-time performance. Third, security: CLD may introduce new security vulnerabilities, requiring the design of corresponding security mechanisms. Fourth, fairness: Ensure that all users have fair access to spectral resources.

4.4. Algorithm Deployment Feasibility Analysis

(1) Computational Complexity. We prove that the proposed CLD algorithm runs in $O(N \cdot M \cdot \log M)$ per iteration, where N is the number of SUs and M is the number of subchannels. This is comparable to the state-of-the-art water-filling algorithm and is therefore compatible with real-time operation on a 1 ms LTE slot basis.

(2) Hardware Platform. We specify that the algorithm can be implemented on a Xilinx Zynq UltraScale+ MPSoC (FPGA+ARM) or an NVIDIA Jetson TX2 embedded GPU.

(3) Signaling Overhead. We quantify the extra cross-layer signaling as 64 bytes per user per frame, which is <0.4% of a 1 ms subframe at 20 MHz bandwidth.

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(4) Robustness. We evaluate performance under ±10% channel estimation error and show <3% degradation in energy efficiency, confirming robustness.

5. Conclusion

This paper has delved deeply into the issue of CLD in CRNs and has proposed a comprehensive CLD framework and corresponding strategies. Through simulation experiments, the significant effects of CLD in improving spectrum utilization, reducing transmission latency, and enhancing the reliability of data transmission have been validated. In particular, the close collaboration between the physical and MAC layers and the synergistic optimization between the network and application layers have provided strong support for the comprehensive enhancement of network performance. The research results indicate that CLD can achieve efficient utilization and management of spectrum resources, meeting the needs of different users and applications, which is of great significance in addressing the challenge of scarce spectrum resources. In addition, this paper has also analyzed the challenges that CLD may face in practical applications, such as the timeliness of information, interlayer communication overhead, and security issues, and has proposed corresponding solutions.

Future research will focus on enhancing the security of CRNs, developing efficient multiobjective optimization algorithms, integrating artificial intelligence and machine learning technologies to improve the intelligence of spectrum sensing and resource management, improving the processing of real-time CSI, and experimentally verifying CLD in real environments. Distinct from prior layer-pair optimizations, this work introduces the first four-layer joint optimization framework with strict latency guarantees.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Funding

This study is funded by the National Natural Science Foundation of China, 10.13039/501100001809, 62261047; the Guizhou Provincial Basic Research Program (Natural Science), Nos. ZK [2023]049 and MS [2025]098; and the Science and Technology Innovation Team of Guizhou Education Department, Grant No. [2023]094.

Acknowledgments

This research was supported by the National Natural Science Foundation of China (62261047), the Guizhou Provincial Basic Research Program (Natural Science) (No. ZK [2023]049 and MS [2025]098), and the Science and Technology Innovation Team of Guizhou Education Department (Grant No. [2023]094).

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